

ROHIT JHA D

Data Scientist | Machine Learning Engineer

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PROFESSIONAL SUMMARY

Data Scientist and Applied Machine Learning Engineer with enterprise predictive modeling experience at HPCL and deep learning research background at C-DAC. Expertise across the end-to-end ML lifecycle: Exploratory Data Analysis (EDA), feature engineering, mathematical modeling, model evaluation (AUC-ROC, RMSE, F1), and production deployment. Proficient in Python (Pandas, NumPy, Scikit-learn), PyTorch, Hugging Face, vector retrieval, and REST API deployment (FastAPI/Docker). Proven track record deploying unsupervised anomaly detection models (reducing false alarms by 40%) and operational forecasting systems across 1,400+ endpoints.

TECHNICAL SKILLS

- **Machine Learning & Algorithms:** Supervised & Unsupervised Learning, Regression, Classification, Random Forest, XGBoost, Scikit-learn, Cross-Validation, Hyperparameter Tuning
- **Deep Learning & Multimodal AI:** PyTorch, TensorFlow, Deep Autoencoders, Isolation Forest, Hugging Face Transformers, CLIP (Vision-Language), Vector Embeddings, Cosine Similarity
- **Data Science & Programming:** Python (NumPy, Pandas, SciPy, Matplotlib, Seaborn), SQL (PostgreSQL, MySQL - Advanced Queries, Data Extraction), R, Elasticsearch
- **Feature Engineering & Statistics:** Exploratory Data Analysis (EDA), Dimensionality Reduction (PCA), Hypothesis Testing, A/B Testing, Probability Distributions, Data Preprocessing
- **MLOps & Production Deployment:** FastAPI, Flask, Docker, RESTful APIs, Git, GitHub, Linux/Bash, Model Monitoring & Latency Optimization, Vector Databases (pgvector, Qdrant)

PROFESSIONAL EXPERIENCE

Centre for Development of Advanced Computing (C-DAC)

Chennai, India | Jan 2023 – Jun 2023

Machine Learning Research Intern – Deep Learning & Anomaly Modeling

- Formulated and implemented unsupervised deep learning architectures (Deep Autoencoders and Isolation Forest) in PyTorch to detect multivariate anomalies across 500,000+ daily log sequences.
- Engineered robust feature transformation pipelines using Python, scaling and extracting latent behavioral embeddings to reduce false discovery rate by 40%.
- Optimized model evaluation metrics, improving anomaly detection recall by 25% on holdout validation data via reconstruction error threshold calibration.
- Benchmarked latent representation learning against classical statistical baselines, documenting model drift characteristics and inference latency for production deployment.

Hindustan Petroleum Corporation Limited (HPCL)

Trichy, India | Apr 2024 – Mar 2025

Data Science & Analytics Trainee – Predictive Operations (NATS Apprenticeship)

- Developed automated ETL pipelines using Python and SQL across SAP ERP, CRIS, and TIBCO Spotfire, standardizing multi-source supply chain datasets for machine learning modeling.
- Formulated statistical risk scoring models across INR 90+ Crore in B2B accounts, predicting default likelihood on overdue receivables (DV >15 days) to safeguard working capital.
- Applied time-series clustering and variance modeling across 1,400+ fuel retail stations to forecast dryout likelihood and optimize supply replenishment schedules.
- Automated analytical computation pipelines and KPI tracking dashboards in Python and SQL, cutting recurring reporting turnaround time by 35% across 5 sales territories.

MACHINE LEARNING & AI PROJECTS

Multimodal AI Semantic Search Engine | PyTorch, OpenAI CLIP, Streamlit, Vector Retrieval

- Engineered a zero-shot cross-modal retrieval system using OpenAI's CLIP architecture to project natural language queries and image assets into a joint 512-dimensional embedding space.
- Implemented high-dimensional dense vector indexing and Cosine Similarity computations, achieving sub-second Top-K nearest-neighbor semantic search retrieval.
- Constructed an end-to-end Python inference pipeline wrapped in a responsive Streamlit UI for real-time natural language querying across diverse datasets.

Vehicle Telemetry Regression & Analytics Service | Scikit-learn, Python, Flask, REST API

- Conducted end-to-end Exploratory Data Analysis (EDA), feature selection, and correlation matrices across multi-sensor engine telemetry records.
- Trained, cross-validated, and hyperparameter-tuned a Random Forest Regressor, achieving an R^2 score of 0.91 with minimized root-mean-square error (RMSE).
- Containerized and deployed the predictive model as a low-latency RESTful inference API using Flask for real-time automated scoring.

EDUCATION

Bachelor of Technology (B.Tech.) in Information Technology

Aug 2019 – Jul 2023

Anna University (Gojan School of Business and Technology), Chennai | CGPA: 8.42 / 10.0

CERTIFICATIONS

- **Google Data Analytics Professional Certificate** – Coursera / Google (SQL, R, Statistical Foundations, Predictive Analytics, EDA)
- **Graduate Apprenticeship Training Completion Certificate (NATS)** – Ministry of Education, Govt. of India (HPCL Trichy)